

Supplementary Distribution Alignment Results

This document reports supplementary means and standard deviations for the alignment between model-generated distributions and observed human data. We provide results on two datasets: the Big-Five Personality dataset (Table ??) and the MobLab dataset (Table D.2), covering personality traits and economic behaviors, respectively.

Table D.1. Means and standard deviations for both the observed and generated data of personality traits in the Big Five test.

Method	Openness	Conscientiousness	Extraversion	Agreeableness	Neuroticism
Observed data $\mathcal{D}_{\text{obs}}$	39.09 ± 6.26	33.47 ± 7.31	30.11 ± 9.22	38.45 ± 7.15	30.97 ± 8.62
1,000 Human Samples	38.97 ± 6.27	33.67 ± 7.38	30.22 ± 9.48	38.53 ± 7.24	30.75 ± 8.55
Fixed System Prompt	44.80 ± 1.56	41.13 ± 1.56	32.56 ± 2.09	47.06 ± 1.26	28.66 ± 2.05
Persona Hub	47.30 ± 1.73	45.07 ± 2.70	33.55 ± 3.94	47.15 ± 2.59	25.70 ± 3.63
OASIS	44.34 ± 2.57	43.34 ± 3.76	38.62 ± 6.76	48.61 ± 1.73	23.56 ± 2.67
Modular Pluralism	39.79 ± 1.92	38.94 ± 2.30	31.00 ± 2.23	43.03 ± 2.22	34.61 ± 2.78
Persona Hub *	41.68 ± 3.19	38.08 ± 3.10	30.42 ± 8.52	38.64 ± 6.85	30.74 ± 8.37
Modular Pluralism *	39.81 ± 1.96	38.90 ± 2.33	30.96 ± 2.30	42.93 ± 2.27	34.57 ± 2.91
Ours (EM-based)	39.43 ± 6.24	33.16 ± 7.10	30.06 ± 8.99	38.59 ± 7.77	30.97 ± 8.35
Ours (GB-based)	38.32 ± 6.10	32.53 ± 6.56	29.66 ± 8.84	38.20 ± 6.94	31.35 ± 8.40

Table D.2. Means and standard deviations for both the observed and generated data of economic behaviors in MobLab games.

Method	Dictator	Proposer	Responder	Investor	Banker	Public Goods	Bomb
Observed data $\mathcal{D}_{\text{obs}}$	25.70 ± 21.73	44.13 ± 19.99	34.90 ± 20.31	42.02 ± 36.07	59.38 ± 39.51	9.63 ± 6.44	45.35 ± 24.67
1,000 Human Samples	25.89 ± 22.15	43.34 ± 20.41	34.78 ± 20.18	42.49 ± 36.55	59.87 ± 39.27	9.50 ± 6.43	46.08 ± 24.27
Fixed System Prompt	49.96 ± 0.89	49.05 ± 3.03	2.18 ± 7.50	64.34 ± 22.65	82.31 ± 11.10	10.04 ± 0.97	46.28 ± 14.12
Persona Hub	50.00 ± 0.00	48.71 ± 3.69	16.97 ± 22.53	71.85 ± 25.00	86.34 ± 12.38	10.30 ± 1.41	43.71 ± 16.61
Modular Pluralism	55.76 ± 10.40	52.67 ± 9.67	30.18 ± 26.12	45.36 ± 33.66	85.95 ± 19.05	10.61 ± 2.54	69.80 ± 41.47
Persona Hub *	50.00 ± 0.00	41.10 ± 11.02	32.74 ± 17.71	42.39 ± 36.58	70.19 ± 23.77	10.57 ± 5.40	45.33 ± 24.58
Modular Pluralism *	50.57 ± 3.77	48.69 ± 10.54	37.20 ± 21.78	43.18 ± 34.24	70.85 ± 25.29	9.63 ± 3.99	43.25 ± 37.00
Ours (EM-based)	24.36 ± 22.08	43.84 ± 20.89	33.25 ± 19.38	42.00 ± 35.63	66.97 ± 32.10	9.43 ± 6.49	41.51 ± 27.62
Ours (GB-based)	26.07 ± 21.90	45.08 ± 20.17	34.52 ± 18.97	42.05 ± 36.09	61.04 ± 37.15	9.71 ± 6.14	45.97 ± 26.89